

E. coli* at Toronto’s Beaches: Are Some Beaches Less Safe Than Others?

Marie Curtis Park East and Sunnyside exceed the swimming limit on a third of summer days, more than seven times as often as Gibraltar Point

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Toronto tests the water at its ten swimming beaches every summer day and posts a warning when *E. coli*, a sign of faecal pollution, exceeds its limit. Using 98,370 results published by the City from 2007 to 2026, this paper compares how often each beach was over the limit and how well a sign based on the previous day’s samples matched the water quality on the day it was posted. Marie Curtis Park East and Sunnyside were over the limit on about a third of days, more than seven times as often as the cleanest beach, whereas the daily sign missed nearly two thirds of the days over the limit. For swimmers, the choice of beach says more about the water quality than the sign posted on that day.

1 Introduction

E. coli is a sign of faecal pollution. It lives in the gut of people and other warm-blooded animals, so finding it in lake water means that sewage, storm-water runoff or bird droppings have arrived recently, bringing the viruses and bacteria that make swimmers ill [1]. Swimming in water above Toronto’s standard raises the risk of ear, eye, nose, throat and skin infections, and swallowed water can cause stomach illness. Young children, older people and people with weakened immune systems are most at risk [2].

Toronto Public Health samples the water at the city’s ten supervised beaches every day of the summer and posts a “Warning” sign when *E. coli* exceeds 100 per 100 mL, half the limit used by Ontario and Health Canada [2]. Every result since 2007 is public [3], yet two questions a swimmer might ask have not been answered from them. First, are some beaches over the limit much more often than others, and does that hold from year to year? Second, because laboratory results take about a day, the sign posted on any day reflects the previous day’s water quality; how often is that sign right about the water quality on the day it is posted? Studies of Toronto’s beaches have linked *E. coli* to weather and river flow

*Code and data are available at: <https://github.com/seandata8/toronto-beaches-ecoli>.

[1] and have evaluated same-day testing at two beaches over one summer [4], but they did not compare the ten beaches with one another over the full record, nor did they measure how often the daily sign matches the day’s water quality.

This paper answers both questions using 98,370 results from 2007 to 2026. Two beaches stand apart: Marie Curtis Park East and Sunnyside were over the limit on 34% and 32% of sampled days, more than seven times the 4% at Gibraltar Point, and in 19 of the 20 years both were over the limit more often than every other beach. Compared on the same days, both carried about 2.2 times the *E. coli* of the average beach. The daily sign, by contrast, was a weak guide: on the days when the water was over the limit, the “Warning” sign was up on only 37% of them, and when the “Warning” sign was up, the water was under the limit on 63% of days.

These findings matter to anyone deciding where and when to swim. The choice of beach says more about the likely water quality than the sign posted on the day, because the sign describes water that has since changed. The two effects also compound: at Marie Curtis Park East and Sunnyside, the water was over the limit on about one day in four even when the “Safe” sign was up, compared with one day in 25 at Gibraltar Point. A swimmer at those two beaches is therefore better guided by the beach’s record than by the sign, and those are the beaches where a faster warning, from same-day testing or a prediction based on the weather, would protect the most people (Section 4).

Section 2 describes the data, how a water sample becomes a result, and the methods. Section 3 compares the beaches and measures how often the daily sign was right. Section 4 considers why the beaches differ, how the warnings could be improved, and the limitations of this analysis. Appendix A describes the data cleaning, and Appendix B checks the method on simulated data.

2 Data

2.1 Source and context

To reduce water-borne illness among swimmers, Toronto Public Health, Toronto Water, the police Marine Unit and the City’s parks division jointly sample the water at the city’s ten supervised beaches every day from Victoria Day to Labour Day [3]. Samples are taken at four to six fixed sites along each beach (Figure 1), and when the laboratory count of *E. coli* at a beach is too high, Toronto Public Health posts signs warning against swimming [5]. The results are published on the City’s open data portal [3,6] and form the data for this paper.

Toronto’s standard of 100 *E. coli* per 100 mL is half that of Ontario and Health Canada: Health Canada raised its guideline from 100 to 200 in 2012, and Ontario follows it, whereas Toronto retained the earlier, stricter value [1,2]. Because 100 is the level at which Toronto posts warnings, it is the limit used throughout this paper.

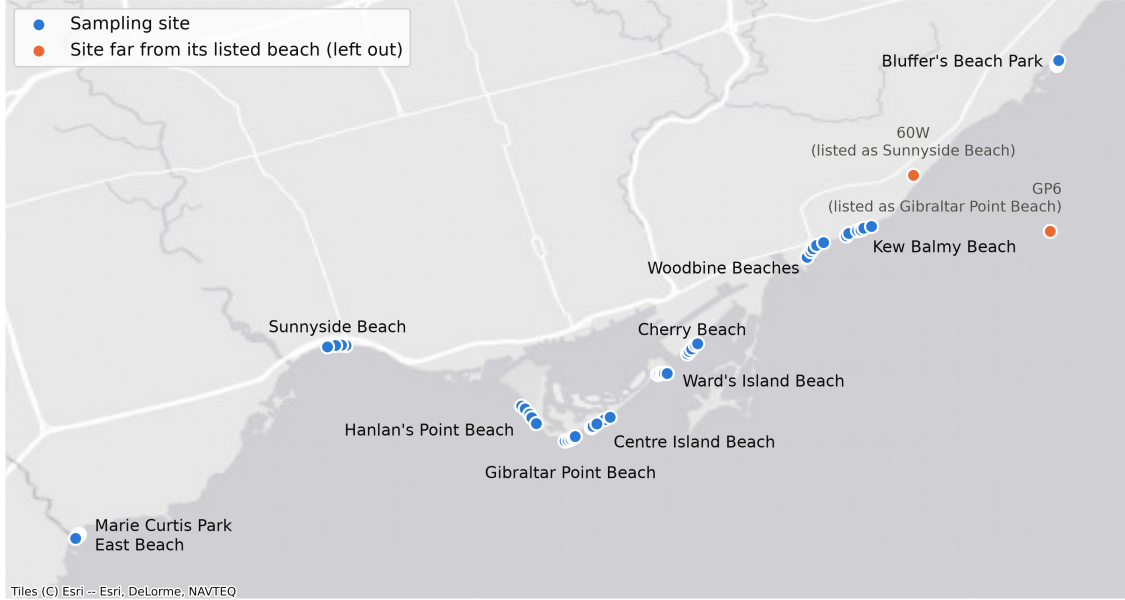


Figure 1. Each of Toronto’s ten swimming beaches is sampled at four to six sites along the shore. Blue points are the 50 sites this paper uses. The two orange points, sites 60W and GP6, were added in 2026 and sit 14 to 16 km from the beaches they are listed under, so they are left out. Map tiles from Esri [7].

Because the data describe water quality rather than people, they raise no privacy concerns; the ethical stakes lie instead in the advice built on them, since a needless warning costs swimmers a beach day and a missed one exposes them to illness [4]. The City also offers each beach’s daily swimming condition for download [5], but the open data were used here because they give every individual result, allowing each day’s status to be calculated the same way for all twenty years.

The analysis uses Python [8] with Polars [9], NumPy [10], statsmodels [11], pointblank [12], matplotlib [13] and contextily [14], and the paper is written in Quarto [15].

2.2 Measurement

Each row of the dataset records one site on one day: the beach, a site code and its location (Figure 1), the date of collection and the *E. coli* result. To obtain each result, staff collect 200 mL of water about 30 cm below the surface at each of the beach’s sites (five at most beaches, six at Kew Balmy and four at Sunnyside), and a Public Health Ontario laboratory counts the *E. coli* by membrane filtration, in which bacteria trapped on a filter grow overnight into visible colonies that are counted and scaled to a volume of 100 mL [16]. Because the process takes about a day, a posted result describes the previous day’s water quality [2,5]. When no sample could be taken, usually because of bad weather, the result was left blank [2]; 3.5% of rows are blank.

The laboratory’s reporting limits compress the data at both ends (Figure 2). Results below 10 are rarely reported, so the 49% of results at exactly 10 are best read as “10 or fewer”. From 2018, high results were increasingly reported as exactly 1,000; earlier results reached 15,820, whereas none exceeded 1,000 in 2026, which suggests that a reporting ceiling was introduced. Because both limits narrow the range, differences among the cleanest beaches and the severity of the worst days are likely understated. The ceiling can also hide a day over the limit: since 2018, 34 days at a beach included a result of 1,000 yet had a daily geometric mean (Section 2.3) below 100, and on about 8 of them a true result of 2,000 or less would have put the day over. The true number of days over the limit since 2018 may therefore be slightly higher than the number calculated from these data.

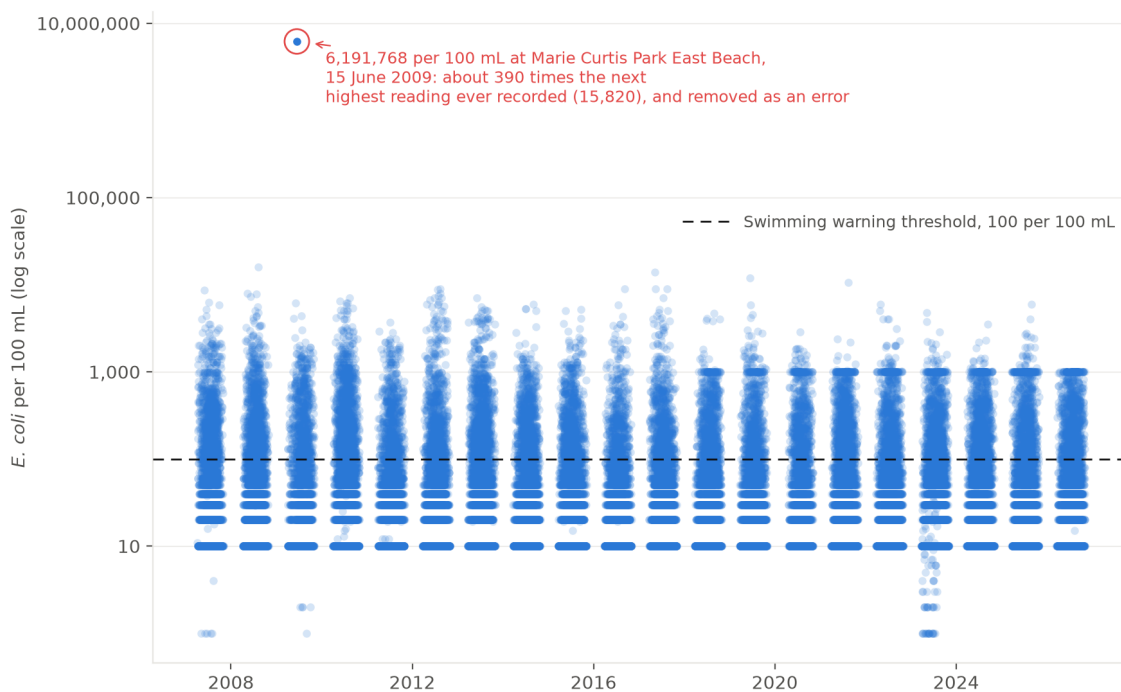


Figure 2. Every result the City has published, 2007 to 2026. One point is one water sample (98,991 in total). The vertical axis is logarithmic, so each gridline is 100 times the one below. Points are spread sideways within each year so that they can be seen; the *E. coli* count itself is never moved. Most results are reported in multiples of 10, which is why the low readings fall into separate rows. The laboratory reports no results as zero, and “no colonies detected” is assumed to be reported as 10. From 2018 on, many results are reported as exactly 1,000, and these are assumed to be readings of 1,000 or more. The dashed line marks 100 *E. coli* per 100 mL, the level at which the City posts a swimming warning. The ringed point is the only reading the analysis removes.

Before analysis, three groups of results were removed: one implausible reading (ringed in Figure 2), the results from two sites that lie far from the beaches they are listed under (Figure 1), and results from dates outside the swimming season. The remaining 98,370

results come from 50 sites on 2,019 days across 20 years; Appendix A gives the details.

Most results are low, but two beaches stand apart (Table 1). At eight of the ten beaches the median result (the middle value when all results are ranked) is 10 or 20, and at Gibraltar Point, the cleanest, 61% of results are at the floor of 10 and only 6% exceed 100. At Marie Curtis Park East and Sunnyside, by contrast, the medians are 50 and 60, only 27% and 20% of results are at the floor, and 35% and 34% exceed 100, about six times the share at Gibraltar Point.

Table 1. Results at each beach, 2007 to 2026. A result is the number of *E. coli* per 100 mL in one water sample. Ten is usually the lowest value the laboratory reports, and 100 is Toronto’s limit for swimming. The 90th percentile is the level that one result in ten exceeds. Beach-days are days with at least four results. Beaches are listed from the highest % above 100 to the lowest.

Beach	Sites	Results	Me- dian	90th percentile	At 10	Above 100	Beach- days
Marie Curtis Park East	5	9,612	50	590	27%	35%	1,925
Sunnyside	4	8,006	60	400	20%	34%	1,986
Centre Island	5	9,645	20	220	40%	19%	1,929
Kew Balmy	6	12,047	20	250	47%	19%	2,011
Bluffer’s Park	5	10,006	10	150	55%	13%	2,001
Woodbine	5	10,035	10	110	59%	11%	2,005
Ward’s Island	5	9,757	10	110	58%	10%	1,952
Cherry	5	10,012	10	90	58%	9%	2,003
Hanlan’s Point	5	9,689	10	80	61%	8%	1,939
Gibraltar Point	5	9,561	10	60	61%	6%	1,911

2.3 Methods

Because Toronto’s limit applies to the geometric mean of a beach’s results on a day rather than to any single result [17], the analysis works with beach-days. The geometric mean of n results is the n th root of their product, or equivalently the average of their logarithms converted back. It suits counts that span orders of magnitude, because a single extreme result does not dominate it: results of 10, 10, 10, 10 and 1,000 give a geometric mean of 25, against an ordinary average of 208. A beach-day was counted only when it had at least four results, rather than five, so that Sunnyside (four sites) is included. This gives 19,662 beach-days, of which 2,816 (14%) were over the limit. A day over the limit is not necessarily a day on which a warning was posted—the data do not record postings. Saleem et al. describe posting decisions based on a geometric mean over two days [4]. In addition, a posted warning reflects the previous day’s water quality, because testing takes about 24 hours (Section 2.2).

To test whether the differences reflect the beaches rather than the weather on the days each was sampled, each beach was compared only with the other beaches sampled on the same

day. For beach b on day d , the model is

$$\log_{10}(G_{bd}) - \bar{L}_d = \beta_b + \varepsilon_{bd}, \quad (1)$$

where, in Equation 1, G_{bd} is the beach’s geometric mean that day, $\bar{L}_d$ is the average of $\log_{10}(G)$ over all beaches sampled that day, β_b is the beach’s average gap from the others, and ε_{bd} is the day-to-day variation the model does not explain. Subtracting each day’s average removes anything that affects all beaches at once, such as a city-wide storm. Because the model works on a log scale, a gap of $\beta_b = 0.3$ means about twice the *E. coli* of the average beach on the same day ($10^{0.3} \approx 2$).

The standard error of each β_b was calculated from the variation between months rather than between days, because a wet week raises a beach’s levels for several days running. The few sampling days in late May and early September were counted with June and August. Grouping days this way roughly doubles the width of the intervals compared with treating every day as independent. Each estimate is reported with a 95% interval: if the analysis were repeated over many different runs of summers, about 95% of such intervals would contain the true gap. Two beaches were said to differ significantly when a gap as large as the one observed would arise less than 5% of the time if they were in fact alike. Testing all 45 pairs this way would produce about two false differences by chance, so the comparisons were corrected by Holm’s method, which raises the bar each comparison must clear so that the chance of any false difference stays near 5%.

Because results take about a day, the sign on display at a beach on any day, called here the posting day, is based on the samples taken the day before. The City posts a “Warning” sign when *E. coli* is over its limit and a “Safe” sign otherwise [2]. To measure how often the sign was right, each posting day was paired with the previous day at the same beach, using only days that were consecutive (18,998 pairs). The sign on the posting day was taken to be “Warning” if the previous day’s geometric mean was over the limit, and “Safe” if it was not. That sign was then compared with the water quality measured on the posting day itself.

3 Results

Two beaches were over the limit far more often than the other eight (Figure 3). Marie Curtis Park East was over the limit on 34% of its sampled days and Sunnyside on 32%, more than seven times the 4% at Gibraltar Point, the cleanest beach. The other seven beaches ranged from 16% at Kew Balmy to 7% at Cherry and Hanlan’s Point. Sunnyside also stands out in the shape of its distribution: it is the only beach with more days between 30 and 100 than below 30, so its typical day is higher, not only its worst. The ranking is not the result of a few bad years. In 19 of the 20 years, both Marie Curtis and Sunnyside were over the limit more often than every other beach; the exception was 2021, when Kew Balmy (27%) passed Sunnyside (25%). Nor is it an effect of the reporting ceiling introduced in 2018: Marie Curtis was over the limit on 37% of days before 2018 and 31% from 2018, Sunnyside on 32% in both periods, and Gibraltar Point on 5% and 4%.

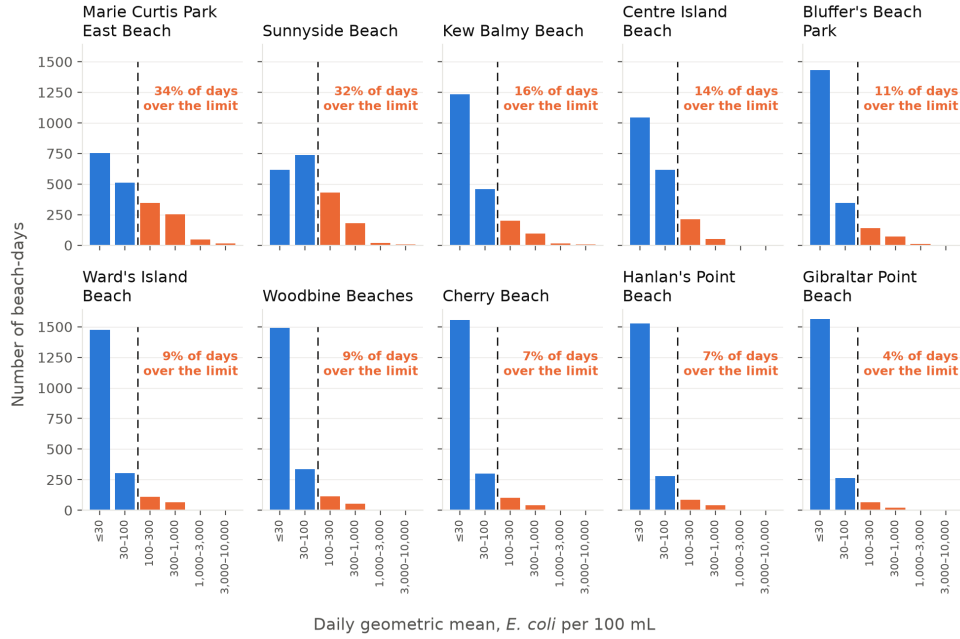


Figure 3. How often each beach went over the limit, 2007 to 2026. Each bar counts the beach-days whose geometric mean falls in that range (19,662 days in total; days with fewer than four results are left out). Orange bars, right of the dashed line, are days above Toronto’s limit of 100 *E. coli* per 100 mL. Panels run from the largest share of days over the limit to the smallest.

The same-day comparison gives the same ranking (Figure 4). On average, Marie Curtis Park East and Sunnyside each had about 2.2 times the *E. coli* of the average beach sampled the same day (95% intervals 2.0 to 2.4 times), whereas Gibraltar Point had about a third less (0.66 times). The two worst beaches did not differ from each other (both 2.19 times), but each differed significantly from every other beach. Of the 45 pairs of beaches, 31 differed after Holm’s correction; the 14 that did not were Marie Curtis and Sunnyside, Centre Island and Kew Balmy, and 12 pairs among the six cleanest beaches. Centre Island ranks above Kew Balmy in the model (1.22 against 1.09 times) but below it in the share of days over the limit (14% against 16%), because the model compares typical levels on the same day, whereas the share counts only how often 100 is crossed. The gaps shown are smaller than the true gaps: in simulated data with known differences, the reporting floor and ceiling reduced the estimated gaps to about 65% of their true size, without reversing any of them (Appendix B).

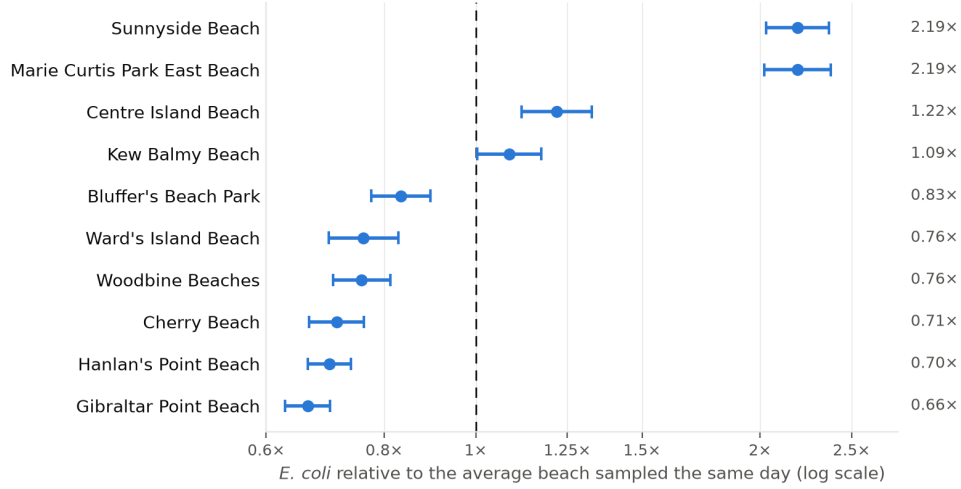


Figure 4. Each beach’s *E. coli*, averaged over all sampled days, as a multiple of the average beach sampled the same day, 2007 to 2026, with 95% intervals. The reporting floor and ceiling make these gaps smaller than the true ones.

Table 2 compares the sign posted at each beach with the water quality measured on the posting day, for 18,998 pairs of consecutive days. The rows give the sign, “Warning” or “Safe”, which is based on the previous day’s samples; the columns give whether the posting day’s water was over or under Toronto’s limit. The sign was right on 1,002 days with a “Warning” sign and water over the limit, and on 14,584 days with a “Safe” sign and water under it. It was wrong on 1,712 days with a “Warning” sign and water under the limit, and on 1,700 days with a “Safe” sign and water over it. Table 3 gives percentages derived from these counts, as answers to questions relevant to a Toronto beach swimmer. The sign was least reliable where it mattered most: under a “Safe” sign, the water was over the limit on 27% of days at Marie Curtis Park East and 25% at Sunnyside, against 4% at Gibraltar Point.

Table 2. Sign posted at each beach, based on the previous day’s samples, against the water quality on the posting day, for pairs of consecutive days, 2007 to 2026. Each cell gives the number of pairs and its share of all pairs.

Sign posted	Water over the limit on the	
	posting day	Water under the limit
“Warning” sign	1,002 (5.3%)	1,712 (9.0%)
“Safe” sign	1,700 (8.9%)	14,584 (76.8%)

Table 3. What the sign says about the water quality on the posting day, 2007 to 2026, derived from the counts in Table 2. The sign is based on the previous day’s samples.

Question	Answer
On any posting day, how often was the water over the limit?	14%
When the “Warning” sign was up, how often was the water under the limit?	63%
When the “Safe” sign was up, how often was the water over the limit?	10%
When the water was over the limit, how often was the “Warning” sign up?	37%

4 Discussion

Twenty years of Toronto’s beach monitoring give a consistent picture. Marie Curtis Park East and Sunnyside were over the limit on about a third of sampled days, more than seven times as often as Gibraltar Point, and this ranking held from year to year and on both sides of the 2018 change in reporting. Compared on the same days, both carried about twice the *E. coli* of the average beach, and because the reporting limits compress the data, the true gaps are likely larger. The sign posted each day, by contrast, was a weak guide to that day’s water quality: it missed nearly two thirds of the days over the limit, and nearly two thirds of the “Warning” signs went up on days when the water was under the limit. For a swimmer, the choice of beach therefore says more about the water quality than the sign posted on the day.

The two worst beaches share a location at the mouth of a river. Marie Curtis Park East lies at the mouth of Etobicoke Creek, and Sunnyside lies beside the mouth of the Humber River, where a breakwall shelters the beach from the waves of the open lake [4]. Both rivers drain urban land, and higher stream flow has been associated with higher *E. coli* at Toronto beaches [1]; at Sunnyside, the breakwall may also hold river water near the shore rather than letting it disperse. By contrast, three of the four island beaches, which face the open lake away from any river mouth, were among the cleanest; Centre Island was the exception. These explanations remain hypotheses that the present analysis does not test. Two analyses of these data could test them: comparing the sites within each beach, since sites nearer a river mouth should have higher results if the river is the source, and linking daily results to river flow and rainfall, to see whether the gap between beaches widens after heavy rain.

The weakness of the daily sign follows from the one-day delay: water quality changes faster than the testing can follow, so a sign based on the previous day’s samples misses most days over the limit and puts up most of its warnings on days under the limit. The City acknowledges this, advising swimmers to stay out of the water for 24 to 48 hours after heavy rain and when the water is cloudy [5]. Two approaches could narrow the gap. The first is a predictive model that estimates the day’s water quality from conditions such as rainfall, wind direction and turbidity; Toronto Public Health piloted one with Environment Canada at Marie Curtis Park East and Sunnyside in 2022, but had suspended it by 2025 [2]. The second is same-day testing. At the same two beaches, a qPCR method, which measures the DNA of *Enterococcus*, another faecal bacterium, within a few hours, indicated

that 5% of the City’s posting decisions had missed a health risk and 23% had closed the beach unnecessarily [4]. Because these are the beaches with the most days over the limit, they are also where a faster or predictive warning would help swimmers most.

Four limitations qualify these results. First, the signs in this paper were reconstructed from the sample results rather than taken from the City’s records of what was posted. Because the City may base its decisions on a geometric mean over two days [4], the signs actually posted may have been right more or less often than the reconstructed ones. Second, the reporting floor at 10 and ceiling at 1,000 compress the data, so the gaps between beaches are understated; in simulated data, only about 65% of a true gap survived. Third, the analysis uses no weather data, so it cannot say how much of the day-to-day variation, or of the difference between beaches, follows from rainfall and wind. Fourth, Sunnyside has four sites where most beaches have five, which makes its daily geometric mean noisier and slightly more likely to cross the limit; in simulated data this added well under one percentage point to its share of days over the limit, far too little to explain its 32%.

The most direct next step is to replace the reconstructed signs with the ones actually posted. The City’s beach water quality page offers each beach’s daily swimming conditions for download [5], which may record the posted sign; comparing those with the water quality measured on each posting day would show how well the real signs performed, and whether a rule based on two days does better than one based on a single day. As suggested above, comparing sites within each beach and linking daily results to river flow and rainfall would then show why the two west-end beaches are worse and whether a weather-based prediction could warn swimmers on the day itself.

In conclusion, twenty years of Toronto’s own monitoring show that where a swimmer goes matters more than the sign posted on the day. Marie Curtis Park East and Sunnyside stand apart from the other eight beaches, and a sign based on the previous day’s samples misses nearly two thirds of the days over the limit. Faster warnings, starting at these two beaches, would protect the most swimmers.

Appendix

A Data cleaning

The data were downloaded from Open Data Toronto on 22 September 2026 [3] and contain 102,534 rows, one for each site on each sampling day from 3 June 2007 to 16 September 2026. Of these, 3,543 (3.5%) have no result, usually because no sample could be taken, which leaves 98,991 results. Three groups of results were then removed. The first is a single reading of 6,191,768 *E. coli* per 100 mL at Marie Curtis Park East on 15 June 2009, about 390 times the next highest result in the record (15,820), which was treated as an error. The second is the 220 results from sites 60W and GP6, both added on 19 May 2026: they are listed under Sunnyside and Gibraltar Point but lie 14 to 16 km away, between Kew Balmy and Bluffer’s Park, whereas every other site lies within 0.4 km of the rest of its beach.¹ The third is 400 results from dates outside the sampling season, taken here as Victoria Day to Labour Day; the City describes the season as June to Labour Day [2], but in most years since 2012 sampling has begun just after Victoria Day. The remaining 98,370 results come from 50 sites on 2,019 days.

No other values were changed. Results at the reporting floor of 10 and at the ceiling of 1,000 were kept as reported, as were the 63 results below 10. For the analysis by day, the 50 beach-days with fewer than four results were left out, leaving 19,662 beach-days. The cleaned data were checked with automated tests written with pointblank [12], covering the columns and their types, the ranges of dates and values, one result per site per day, and the number of sites at each beach.

B Checking the method on simulated data

To confirm that the analysis finds real differences and does not invent them, it was first run on simulated data in which the answer is known. The simulation reproduces the structure of the real data: the same ten beaches and numbers of sites, daily sampling from Victoria Day to Labour Day, 2007 to 2026, and results built on the log scale from a beach baseline, a city-wide daily effect shared by all beaches, a daily effect for each beach, and noise for each sample. Both daily effects carry half of the previous day’s value into the next, so that wet spells last several days. Six beaches were built at a common level, two were raised by 0.2 on the log10 scale (about 1.6 times) and two by 0.4 (about 2.5 times). The simulated results were then reported the way the laboratory reports real ones: rounded to multiples of 10, with nothing below 10, and, from 2018, most results above 1,000 reported as 1,000.

The analysis recovered what was built in (Table 4). Without the reporting floor and ceiling, the estimated gaps were 0.399 and 0.208, against true values of 0.400 and 0.200. With them, the estimates fell to 0.260 and 0.126, about 65% of the true gaps, while the order of the

¹The author asked Toronto Public Health about these two sites by email on 22 September 2026. On 25 September 2026, staff replied that the listing had been referred for correction (personal communication). This paper uses the data as downloaded on 22 September 2026, so a later correction does not change its results, but a new download may list these sites differently.

groups was unchanged. All 12 comparisons between a beach raised by 0.4 and a beach at the common level were found to differ significantly, and none of the 15 comparisons between beaches built at the same level was, so Holm’s correction kept false differences in check. Sunnyside and Kew Balmy, with four and six sites, were both built at the common level, so their shares of days over the limit show the effect of the number of sites alone: at 4.2% and 4.0%, both fell within the range of the other common beaches (4.4% to 6.2%).

Table 4. Gaps above the six beaches built at a common level, on the log10 scale, in simulated data. The reporting floor and ceiling shrink both gaps to about 65% of their true size without changing their order.

	Beaches raised by 0.4	Beaches raised by 0.2
Built into the simulation	0.400	0.200
Estimated, without the floor and ceiling	0.399	0.208
Estimated, with the floor and ceiling	0.260	0.126

References

1. Sanchez J, Tustin J, Heasley C, Patel M, Kelly J, Habjan A, et al. Region-Specific Associations between Environmental Factors and *Escherichia coli* in Freshwater Beaches in Toronto and Niagara Region, Canada. *International Journal of Environmental Research and Public Health*. 2021;18(23):12841. doi:10.3390/ijerph182312841
2. City of Toronto. About Beach Water Quality [Internet]. 2025 [cited 2026 Sep 22]. Available from: <https://www.toronto.ca/community-people/health-wellness-care/health-inspections-monitoring/swimsafe/beach-water-quality/about-beach-water-quality/>
3. Toronto Public Health. Toronto Beaches Water Quality [Internet]. Open Data Toronto; 2026 [cited 2026 Sep 22]. Available from: <https://open.toronto.ca/dataset/toronto-beaches-water-quality/>
4. Saleem F, Schellhorn HE, Simhon A, Edge TA. Same-day *Enterococcus* qPCR results of recreational water quality at two Toronto beaches provide added public health protection and reduced beach days lost. *Canadian Journal of Public Health*. 2023;114:676–87. doi:10.17269/s41997-023-00763-8
5. City of Toronto. Beach Water Quality [Internet]. 2026 [cited 2026 Sep 22]. Available from: <https://www.toronto.ca/community-people/health-wellness-care/health-inspections-monitoring/swimsafe/beach-water-quality/>
6. City of Toronto. Open Data Toronto [Internet]. 2026 [cited 2026 Sep 22]. Available from: <https://open.toronto.ca/>
7. Esri. World Light Gray Canvas Base [Internet]. 2026 [cited 2026 Sep 26]. Available from: https://server.arcgisonline.com/ArcGIS/rest/services/Canvas/World_Light_Gray_Base/MapServer
8. Python Software Foundation. Python: A Programming Language [Internet]. 2026. Available from: <https://www.python.org/>
9. Vink R et al. Polars: Blazingly Fast DataFrames [Internet]. 2026. Available from: <https://pola.rs/>
10. Harris CR, Millman KJ, Walt SJ van der, Gommers R, Virtanen P, Cournapeau D, et al. Array programming with NumPy. *Nature*. 2020;585(7825):357–62. doi:10.1038/s41586-020-2649-2
11. Seabold S, Perktold J. statsmodels: Econometric and Statistical Modeling with Python. In: *Proceedings of the 9th python in science conference*. 2010. p. 92–6. doi:10.25080/Majora-92bf1922-011

12. Iannone R et al. pointblank: Data Validation in Python [Internet]. 2026. Available from: <https://posit-dev.github.io/pointblank/>
13. Hunter JD. Matplotlib: A 2D Graphics Environment. Computing in Science & Engineering. 2007;9(3):90–5. doi:[10.1109/MCSE.2007.55](https://doi.org/10.1109/MCSE.2007.55)
14. Arribas-Bel D et al. contextily: Context Geo-Tiles in Python [Internet]. 2026. Available from: <https://contextily.readthedocs.io/>
15. Allaire JJ, Teague C, Scheidegger C, Xie Y, Dervieux C. Quarto [Internet]. 2026. Available from: <https://quarto.org/>
16. Public Health Ontario. Public Beach Water [Internet]. 2024 [cited 2026 Sep 22]. Available from: <https://www.publichealthontario.ca/en/Laboratory-Services/Test-Information-Index/Public-Beach-Water>
17. Young I, Sanchez JJ, Desta BN, Heasley C, Tustin J. Recreational water exposures and illness outcomes at a freshwater beach in Toronto, Canada: A prospective cohort pilot study. PLOS ONE. 2023;18(6):e0286584. doi:[10.1371/journal.pone.0286584](https://doi.org/10.1371/journal.pone.0286584)